

RALPH WENZ FIELD, WY, USA

WMO: 720345

Lat: 42.796N

Lon: 109.807W

Elev: 7086

StdP: 11.30

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94086

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-10.9	-22.4	2.1	-15.5	-16.7	2.9	-10.0	28.8	20.9	26.2	21.8	4.1	300	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.5	85.8	54.3	81.9	52.9	80.1	52.3	57.7	76.1	55.8	73.1	54.4	72.6	10.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
53.3	78.8	60.6	50.4	70.6	58.2	47.8	64.1	58.1	28.2	77.2	26.7	72.4	25.7	71.8	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	21.1	18.7	-27.0	88.3	5.3	2.1	-30.8	89.8	-33.9	91.0	-36.8	92.1	-40.6	93.6		
			-26.5	59.9	5.9	2.5	-30.7	61.7	-34.2	63.2	-37.5	64.6	-41.7	66.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	14.2	16.3	26.9	36.5	44.9	54.6	62.8	59.0	51.3	39.3	27.1	15.0
	DBStd	18.45	9.92	9.70	9.10	6.82	6.50	6.44	5.04	4.23	5.88	7.35	9.22	10.43
	HDD50	5552	1110	943	717	406	182	30	1	2	55	336	686	1084
	HDD65	10100	1575	1363	1182	855	624	316	102	188	412	798	1136	1549
	CDD50	966	0	0	0	1	23	167	396	282	93	4	0	0
	CDD65	38	0	0	0	0	0	3	33	2	0	0	0	0
	CDH74	2498	0	0	0	0	9	356	1332	666	132	3	0	0
	CDH80	571	0	0	0	0	0	76	395	91	9	0	0	0
Wind	WSAvg	6.8	6.2	6.3	6.9	8.3	7.5	7.5	6.9	6.6	6.5	6.8	6.6	5.8
Precipitation	PrecAvg	11.5	0.6	0.6	0.6	0.9	1.7	1.3	1.1	1.1	1.2	0.8	0.7	0.6
	PrecMax	19.4	1.9	1.5	1.5	2.4	4.1	3.9	3.0	3.1	3.5	2.3	2.0	1.9
	PrecMin	5.2	0.0	0.1	0.2	0.0	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.0	0.4	0.4	0.4	0.5	1.1	1.0	0.7	0.6	0.7	0.6	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.0	42.7	58.6	67.9	75.0	86.5	93.0	85.8	81.3	71.3	56.9	43.0
		MCWB	32.1	33.9	41.1	44.8	47.7	53.8	57.7	52.7	52.0	47.6	40.5	34.3
	2%	DB	34.5	36.7	51.6	61.3	72.2	81.5	88.0	82.1	77.6	63.7	52.3	36.6
		MCWB	30.5	32.2	38.2	41.6	47.3	52.0	55.8	51.8	49.5	44.7	38.8	31.8
	5%	DB	32.2	34.1	45.2	56.9	68.1	78.7	84.0	80.8	74.7	60.5	46.2	34.0
		MCWB	29.1	30.6	36.0	39.9	46.2	51.0	54.7	51.8	48.7	43.1	36.3	30.6
	10%	DB	30.0	32.0	40.9	52.6	63.1	73.4	81.2	78.7	71.8	55.1	41.4	31.6
		MCWB	27.4	29.1	33.7	38.9	44.8	49.7	53.4	51.5	47.6	41.1	33.6	29.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.4	35.5	42.1	45.8	50.3	55.8	64.2	57.3	55.7	49.5	43.0	35.4
		MCDB	35.4	39.9	55.3	64.9	67.8	76.6	73.3	69.5	70.4	61.4	55.7	38.9
	2%	WB	31.6	33.3	39.1	43.1	48.7	54.1	60.0	55.5	53.4	47.1	39.7	32.8
		MCDB	33.6	36.4	49.4	57.7	66.6	74.8	79.7	70.1	67.1	58.8	48.9	35.3
	5%	WB	29.4	31.3	36.7	41.4	47.3	52.7	57.3	54.2	51.3	45.0	37.5	30.8
		MCDB	31.5	33.6	44.1	54.1	64.2	72.7	77.0	71.5	66.3	56.3	45.0	33.4
	10%	WB	27.3	29.1	34.2	39.4	45.8	51.2	55.6	53.1	49.7	42.6	34.6	28.3
		MCDB	29.6	31.2	39.6	51.3	61.0	70.0	73.5	72.1	65.8	53.6	41.0	30.7
Mean Daily Temperature Range	5% DB	MDBR	25.0	24.5	22.8	23.9	27.2	32.2	35.5	35.8	33.7	26.6	24.4	23.9
		MCDBR	23.8	22.0	25.9	34.0	38.0	40.0	40.0	40.9	41.7	35.6	29.6	21.6
		MCWBR	19.9	17.7	15.0	16.1	16.2	14.0	12.5	15.1	16.8	17.8	17.8	16.9
	5% WB	MCDBR	22.1	20.3	23.8	29.2	33.7	34.9	32.6	33.2	33.0	30.0	26.0	20.2
		MCWBR	19.3	16.8	14.3	15.0	15.1	13.2	11.0	12.7	14.3	15.7	15.8	16.6
Clear-Sky Solar Irradiance	taub	0.225	0.235	0.248	0.291	0.304	0.304	0.326	0.320	0.290	0.260	0.236	0.223	
	taud	2.399	2.380	2.445	2.404	2.466	2.506	2.466	2.475	2.554	2.608	2.555	2.453	
	Ebn at Noon	300	313	320	311	307	306	298	297	301	302	296	292	
	Edh at Noon	24	29	31	36	34	33	34	33	28	23	20	21	
All-Sky Solar Radiation	RadAvg	671	961	1380	1722	1928	2312	2286	1978	1595	1071	706	540	
	RadStd	43	83	89	107	164	175	118	84	126	78	56	42	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page